

Improving Resiliency for Large-Scale Water Suppliers Facing Drought-Related Financial Risk

Madison Haley
Center on Financial Risk in
Environmental Systems
UNC at Chapel Hill
Chapel Hill, NC, USA
mchaley@unc.edu

David Gorelick, PhD
Center on Financial Risk in
Environmental Systems
UNC at Chapel Hill
Chapel Hill, NC, USA
degorelick@unc.edu

Gregory Characklis, PhD
Center on Financial Risk in
Environmental Systems
UNC at Chapel Hill
Chapel Hill, NC, USA
charack@unc.edu

Rachel Kleiman
Center on Financial Risk in
Environmental Systems
UNC at Chapel Hill
Chapel Hill, NC, USA
rkleiman@unc.edu

Juliet Homer
Pacific Northwest National
Laboratory
Richland, WA, USA
juliet.homer@pnnl.gov

Konstantinos Oikonomou, PhD
Pacific Northwest National
Laboratory
Richland, WA, USA
konstantinos.oikonomou@pnnl.gov

Abstract—Hydrologic variability presents significant challenges to water providers in terms of managing supply risk, but resilient systems must also manage the corresponding financial risk. During drought water providers pump less water, make fewer water deliveries, and earn less revenue. Meanwhile, their costs are mostly fixed, with a large fraction linked to debt service resulting from the capital-intensive development of water supply infrastructure. The consequent budget shortfalls can be very disruptive and must be managed effectively if this critical infrastructure is to remain resilient in a future that involves increasingly frequent and severe droughts. For this research, an object-oriented model was developed that integrates hydrologic and financial systems to provide new insights into ways in which large water providers can better quantify and manage their financial risk and establish strategies for promoting greater financial resiliency in the face of climate change. This work involves a case study of the Central Arizona Project (CAP), one of the largest water providers in the Western U.S. that serves nearly 6 million people (e.g., Phoenix, Tucson), as well as large tracts of irrigated land and the demands of several Native American tribes. Current reductions in CAP's access to Colorado River water will reduce its water deliveries by 40% this year, resulting in substantially lower revenues. In response, CAP will raise its wholesale water prices significantly so that it can meet its costs. As an alternative to increasing wholesale prices, CAP could consider reducing the financial risks of hydrologic uncertainty by utilizing some portion of its reserve funds or leveraging its ad valorem tax. Results suggest that CAP's use of reserves could limit annual price increases to 10% or 15% per year through 2028, but to limit annual price increases to 5% under drier conditions would nearly deplete existing reserves in a few years.

Keywords—water supply systems, climate change, financial risk, risk management

I. INTRODUCTION

Climate change is increasing the frequency and severity of extreme weather events (e.g., droughts, heatwaves), presenting a host of resilience-based challenges to operators of critical infrastructure (IPCC, 2022). These challenges often involve managing the risk of supply disruptions of vital commodities, such as energy and water, but the financial risks to these

operators have also been attracting increasing attention from both public- and private-sector actors (The White House, 2021; FitchRatings, 2023; BlackRock, 2021). Large-scale water providers, including water utilities and irrigation districts, build and operate the infrastructure that deliver water to the vast majority of municipal and agricultural water users in the U.S., while also being some of the largest consumers of electricity (e.g., pumping) (NYSERDA, 2010; Chaudhry and Shrier, 2010). Drought and heatwaves, whose occurrence is often correlated, present not only supply-related risks to these entities, but also financial risks, which if not effectively managed will lead to service disruptions and potentially severe impacts on the communities they serve (Mukherjee and Mishra, 2021; Baum and Characklis, 2020). Quantifying the climate-related financial risks faced by large-scale water providers, and developing strategies for managing this risk will be key to promoting greater resilience in the face of climate extremes.

For most water providers, the majority of revenues are collected from the volumetric sale of water (e.g., \$/m³), meaning that reductions in water availability, and hence water deliveries, can directly reduce revenues (Hanemann, 1997; Hughes et al., 2014). Meanwhile, most of the costs borne by these organizations are typically fixed, with the majority attributable to debt-service arising from financing the extensive and expensive infrastructure they must develop and maintain (Hanemann, 1997). As such, these costs, as well as those associated with long-term contractual obligations, remain largely constant regardless of the amount of water pumped (Hughes et al., 2014). One exception is the cost of electricity, which often makes up 10-15% of annual costs, but which can increase several-fold during particularly hot or cold periods when the price of electricity rises (Cherchi et al., 2015).

Drought-induced reductions in water availability could therefore result in intermittent revenue reductions that are disruptive to the traditional cost recovery budgetary model of most water providers in which water rates (i.e. prices) are typically set such that projected revenues meet expected costs (Hughes et al., 2014; Tiger et al., 2014). These challenges are exacerbated by the inability of most water providers, which are often public or pseudo-public organizations, to quickly increase prices due to legal and/or political constraints (Leurig, 2010;

Olmstead and Stavins, 2009). Addressing these risks is critical for water utilities that provide essential services to their communities; if they cannot manage the financial risks borne from environmental hazards and continue to provide affordable services, people's well-being and a community's economic viability are at risk (Denilenko, et al. 2010). The intermittency and variable severity of drought make it one of the most difficult financial risks for water utilities (Hughes et al., 2014).

While communities across the world have long struggled with the management of variable water supplies, current climate change projections predict that extreme hydrologic events are likely to become more frequent and severe in the future (IPCC, 2022). This has been of particular concern in the western United States, where there is a constant struggle to meet municipal and agricultural water demands with available water supplies (Pint, 1999). Historically, states and utilities have hedged against periods of water scarcity and natural fluctuations in rainfall/streamflow via redundancies in supply infrastructure, often in the form of large reservoirs with the capacity to meet demands even during severe drought (e.g. Lake Mead and Lake Powell on the Colorado River) (Gleick, 2003; Kundzewicz et al., 2007). However, as demands have continued to grow and hydrologic variability increased, this capacity is often no longer enough to manage the worst droughts, meanwhile developing new supplies has become much more expensive and difficult to permit (e.g., regulatory approval) (Brajer et al., 1989; Leurig, 2010). This increasing mismatch has obvious implications for the supply risk faced by large water providers, but the financial risks have begun to receive increasing attention.

II. BACKGROUND

The potential for drought-related financial instability has recently become apparent to the Central Arizona Project (CAP), the largest provider of raw (i.e. untreated) water in Arizona and one of the largest in the U.S. The CAP operates a large infrastructure network that pumps water from the Colorado River to over 80% of Arizona's population and nearly 1 million acres of irrigated farmland (CAP, 2021; USBR). Due to the recent megadrought in the Colorado River basin, the worst in over 1,200 years, the volume of water allocated to Arizona through the Colorado River Compact, which governs water use amongst the seven states in the basin, has been reduced roughly 30% from 2020 to 2023, and 40% from its maximum allocation (CAP 2022; Williams et al., 2022). This reduces the amount that CAP pumps to its contracting partners (e.g., urban water utilities, irrigation districts, tribal users) and the revenue earned by CAP from these deliveries (CAP 2021; CAP 2022). Consequently, these cuts to annual pumping have driven CAP to raise its volumetric delivery charges over 40% since 2020 (e.g., from \$155 per acre-foot to \$217 per acre-foot for municipal customers), a price rise that has generated unprecedented budgetary pressures on CAP customers, including the cities of Phoenix and Tucson (CAP, 2021; CAP, 2022).

One alternative that would allow CAP to reduce the size of these rate increases, as well as those it will likely need to impose for at least the next few years (given that Lake Mead levels, which dictate CAP allocation, are unlikely to recover in the near-term), would be using some portion of its reserve funds. The CAP maintains several reserves, as do most large water providers, and could use these to offset some portion of rate

increases. At CAP, relevant reserve funds include a rate stabilization fund, which is nearly depleted and hasn't been funded since 2018, and an Extraordinary Cost Reserve Fund, the use of which is subject to CAP Board approval and is largely funded by tax revenues. While CAP maintains these reserves, the tradeoffs between price increases and reserve fund withdrawals as a sustainable management strategy have not yet been explored. In addition, while power price uncertainty is a secondary financial risk, it may still be material. Given the correlation between drought and extreme temperatures, this gives rise to unanswered questions about the potential compounding of financial risks through a combination of water supply-driven revenue reductions and temperature-related increases in electricity costs (which account for 30% of CAP's total costs in an average year).

While prior literature has explored the financial risks that water providers experience as a result of hydrologic variability (Baum and Characklis, 2020; Baum et al., 2018; Zeff and Characklis, 2013), the scale of CAP and its drought-related risks, as well as its potential vulnerability to electricity price volatility, suggest that it and/or its customers will experience substantial financial risks that will need to be managed more effectively. This research seeks to use an object-oriented modeling approach to provide new insights into ways in which large water providers operating large-scale critical infrastructure can better quantify and manage their financial risks and establish strategies for promoting greater financial resiliency in the face of climate change. The results of this investigation should be of interest to other operators of large-scale water infrastructure (e.g., State (CA) Water Project, Central Valley Project), as well as other groups with similar exposure to joint water-power supply financial risks, including the Power Marketing Administrations (e.g., Western Area Power Administration).

The Central Arizona Project (CAP) supplies up to 1.6 million acre-feet per year (2.0 million megaliters) of raw (untreated) water from the Colorado River to approximately 60 municipal, industrial, agricultural, and tribal customers. It does so by maintaining and operating a complex system of pumps, aqueducts and reservoirs that withdraw water from Lake Havasu and move it over 330 miles to its terminus near the City of Tucson (Figure 1) (CAP, 2021). The CAP was originally commissioned to supply water to growing communities across Arizona by making use of the state's previously underutilized portion of its rights to the Colorado River. Since its completion in 1992, the CAP has become the primary water supply for Arizona's major population centers (Phoenix and Tucson), supporting nearly 6 million people as well as the demands of nearly 1 million acres of irrigated land and the cultural and economic demands of several Native American tribes (CAP, 2021; USBR n.d.).

Water rights to the Colorado River are complex and divided among the seven basin states (CO, NM, UT, WY, CA, NV, and AZ). In a typical year, the state of Arizona is allocated a total of 2.6 million acre-ft, ~60% of which typically flows through the Central Arizona Project while the remainder is used mostly by irrigators along the Colorado River. During severe drought, however, this allocation can be cut, with the size of the cut determined by water levels in Lake Mead, the Colorado's largest reservoir. Cutbacks in Colorado River allocations are enacted via a tiered system described by the Drought Contingency Plan

(DCP), which was activated for the first time in the history of the Colorado system in 2020 in response to the current unprecedented drought conditions (CAP, 2021).



Fig. 1. Map of the Central Arizona Project

Reductions in allocation to the lower basin states (AZ, CA, NV) begin at tier 0 when Lake Mead level drops below 1,090 feet above mean sea level (msl), and further cuts are enforced at successively lower water levels (CAP, 2021). In 2022, the level in Lake Mead dropped below 1,050 msl for the first time and a tier 2a shortage was declared, reducing CAP’s water allocation nearly 40% relative to its full allocation. These reductions affect CAP customers differently as there is also a tiered priority system for CAP water delivery, with agriculture typically receiving the first cuts and, as drought conditions worsen, municipal & industrial and tribal customers beginning to have their allocations reduced in equal proportion. In tier 2a of the DCP, CAP’s allocation is reduced from a maximum of 1.6 MAF to approximately 1.0 MAF, with the vast majority of this reduction absorbed by irrigators. The resulting reduction in CAP water availability and the amount delivered to its customers has obvious implications for water supply, but also presents CAP with some unprecedented financial challenges.

Like most water utilities, the majority of CAP’s revenues come in the form of volumetric water delivery fees (priced in \$ per acre-foot) that are designed to recoup both CAP’s operations and maintenance (O&M) costs and capital charges (e.g., debt service). A reduction in water deliveries will have a direct impact on CAP revenues, although the size of the impact depends on whose allocation is being curtailed, as different users pay different fees, with M&I being charged the highest (Figure 5). In addition to volumetric charges, CAP collects revenues through an ad valorem tax applied to property owners in its service area, fees associated with groundwater storage and management, and various other small assessments (Figure 2).

While the majority of revenues are subject to risk when water deliveries are reduced, the majority of CAP’s costs, approximately 68%, are fixed (or at least difficult to change within a one to two year window consistent with drought), including salaries, payments on debt, and fixed operating costs (Figure 2). There is, however, one part of CAP’s cost structure that is subject to considerable potential variability, and that is the electricity costs associated with pumping and transmission, which together account for a little more than 30% of total costs.

Nonetheless, CAP has a relatively sophisticated approach to managing variability in electricity prices (which can be significant), so while this remains a potential financial vulnerability, it is likely to be secondary to that which CAP faces from drought (CAP, n.d.).

To compensate for revenue lost due to drought and the reduced water availability and deliveries that result, CAP has raised its rates 20% in 2022 and another 13% in 2023. Further rate increases, perhaps significantly larger, will likely need to continue for at least the next few years as Lake Mead levels are projected to remain below the DCP thresholds. The next section describes the approach used for both evaluating CAP’s future financial risk over the period 2023-2028 and several strategies for managing that risk.

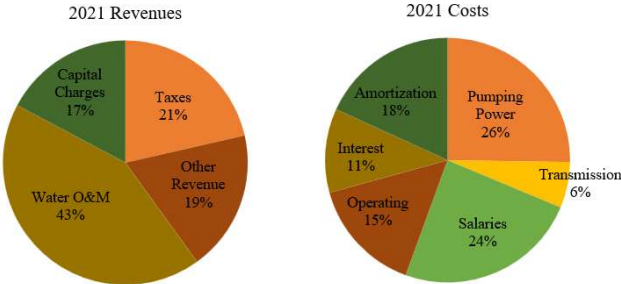


Fig. 2. Central Arizona Project 2021 Budget Revenues and Costs

III. METHODS

In order to quantify CAP’s financial risk and evaluate strategies for managing it, an object-oriented model has been developed in Python to simulate the coupled hydrologic, engineered, financial, and institutional systems at a monthly resolution that have bearing on its financial exposure. The general modeling platform is described in Figure 3.

A. Physical System and Water Allocations

The model uses the elevation of Lake Mead to determine water supply allocated to CAP each year. Monthly time-series of Lake Mead levels under 32 different hydrologic futures have been developed by the Bureau of Reclamation using the Colorado River Mid-term Modeling System (CRMMS). These hydrologic scenarios have been largely accepted by Colorado River users as reflective of the envelope of potential futures, but for the sake of simplicity, this analysis will focus on only three which individually represent a wet, mid (i.e. average), and dry scenario over the simulation period (Figure 4).

Annual allocations to the Central Arizona Project are made in accordance with existing water rights laws and agreements, including the Colorado River Compact, Drought Contingency Plan (DCP) and Intentionally Created Surplus (ICS) agreement. The total CAP allocation is then divided among competing priority classes, including M&I, Federal Contract (Tribal), Agricultural Settlement Pool (Ag Pool), and Excess (which, when available, is permitted to specific agricultural customers), each of which is charged a different rate. Each customer in the CAP system, including the cities of Phoenix and Tucson, acquires water via CAP each year in line with their water rights, but can also temporarily acquire water via leases from other CAP water rights holders. Within the model, the object-oriented modeling approach allows water to be allocated in a similar way, in other words, based on institutional rules and existing leases,

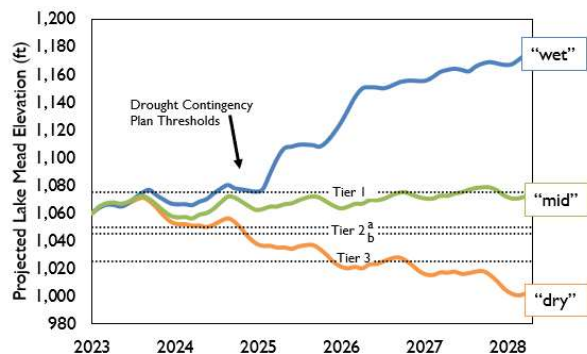
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graph TD
    A[Lake Mead Elevation] --> B(CO River Supply for CAP)
    B --> C(Deliverable water by priority class)
    D[Existing CAP Contractor Entitlements] --> C
    C --> E(Projected Monthly Deliveries by user, by priority, by location)
    F[Historic Delivery and Request Trends] --> E
    G[Lake Pleasant Operations and Pumping] --> E
    H[Projected Power Prices and Uncertainty] --> G
    E --> I(Actual Monthly Deliveries by user, by priority, by location)
    J[Leases and Exchanges among Contractors] --> I
    I --> K(Deliveries to GW Facilities)
    L[Reserve Fund Use] --> K
    K --> M(Annual Budget)
    N[Budgeted Water Delivery Rates] --> M
    O[Historic Pumping Station Efficiency] --> M
    P[Historic Power Prices] --> M
    
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Legend

- Model Calculation (Blue Oval)
- Input Data (Red Rectangle)
- Model Calculation (Dashed Red Oval)

The model also simulates each aspect of infrastructure, including pumping stations, Lake Pleasant (the largest reservoir in the CAP system), and the active management areas (AMAs, groundwater storage facilities). Lake Pleasant storage enables CAP to pump water into the lake during off-peak times when electricity is cheap (spring and at night) and release water for electricity generation during peak price periods (i.e. summer). The model is sufficiently resolved at both the temporal and spatial scales that it captures these changes in operations and tracks deliveries to over 60 individual customers, allowing for a detailed analysis of the allocation of water across CAP's system and the impacts of droughts of varying severity.



B. Financial System

Costs are divided into fixed and variable, with fixed costs (e.g., debt service, salaries) remaining constant in the model runs considered here. Variable costs, including pumping power and transmission costs, are dependent on the quantity of water being pumped. The CAP purchases electricity in three ways that are

vulnerable to market forces (the day ahead market, monthly contracts, and power purchase agreements (PPAs) with the Salt River Project (SRP)) and two that are considered more stable (PPAs from solar producers and the Hoover Dam). CAP has a relatively sophisticated approach to balancing each of these energy sources and managing variability in the price of electricity. While modeling this complexity will become the focus of future work, for this preliminary analysis a simplified regression was used to approximate the cost of energy as a function of water allocation.

Table 1. Rate Components Paid by each “Type” of Customer

“Type” of Customer	Capital Charge	Fixed OM&R	Pumping Energy Rate	Property Tax Equiv.
Municipal and Industrial	•	•	•	
Tribal Contracts		•	•	
Agricultural Settlement Pool			•	
Excess	•	•	•	
Interstate	•	•	•	•

The component of rate revenues that are most vulnerable to drought and reduced CAP allocations is OM&R (operations, maintenance, and replacement). All customers, aside from the agricultural settlement pool, are required to pay this portion of the water rate, and it constitutes the majority of each customer's water bill. The rate charged for OM&R is generally the cost of operating CAP divided by the quantity of water delivered; so, as water deliveries fall, this component of the rate will rise accordingly.

IV. RESULTS

The Central Arizona Project is given the authority to raise and lower water rates each year to recover its costs, meaning that there are no rules limiting how rapidly rates can rise. While this is a financially sustainable financial system for CAP, this transfers the financial risk of drought to its customers, which are often municipalities and water utilities themselves. Given that municipalities typically set their budgets years in advance and are unable to raise additional revenues quickly, these cities are unable to adapt rapidly to these unexpected costs. This has had a significant effect for CAP customers, where cities like

Gilbert and Mesa planned to forego critical infrastructure maintenance in 2022 to pay raw water bills (CAP, 2022).

Concerns over drought-related rate increases are due to both their size and their unpredictable timing. Thus, CAP's customers could benefit from CAP adopting a risk management strategy that gave them greater certainty of smaller annual increases as opposed to larger, unpredictable ones. One strategy to manage this involves the use of reserves to offset some portion of the required rate increases to make them both smaller and more predictable. While CAP maintains reserve funds, most have never been considered as part of a risk management strategy (the small rate stabilization fund aside). To explore this possibility, the model was used to project future rate increases under the considered supply scenarios, with a "no rate stabilization" baseline representing future rates under the current rate structure (Figure 5). Next, annual rate increase limits were imposed to limit yearly increases to 5%, 10%, and 15%. While figure 5 represents these values for the "mid" hydrologic scenario, the same procedure is also conducted for the "dry" and "wet" scenarios. The difference between the "no rate stabilization" rates and the "rate increase limit" rates represents a revenue shortfall that CAP would have to compensate for with reserve funds. The CAP does, in fact, have several different reserve funds, two of which are relevant in the discussion regarding managing rates. One of these, the "Rate Stabilization" reserve, was established for just this purpose, but the balance is quite small relative to the funds required. The CAP also maintains an "Extraordinary Cost" reserve for managing fluctuations in costs, but its use to manage drought-related rate increases would be unprecedented. In addition to these distinct reserve funds, CAP could leverage its ad valorem tax for rate stabilization.

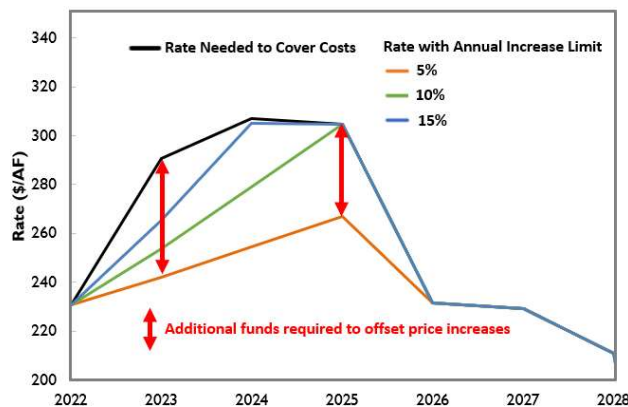


Fig. 5. Projected M&I Rate Fluctuations: Mid-Supply Scenario

The planning period considered here is the present (2022) to 2028, a period over which CAP expects to still be experiencing at least some of the impacts of the current drought. In order to estimate the total amount of supplemental funding required to limit annual rate increases to the 5%, 10% and 15%, the difference between the annual rates at each level and the "no rate stabilization" rates in each year are multiplied times the volume of water CAP delivers. Figure 5 represents these values

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for the "mid" hydrologic scenario, but the same procedure is also conducted for the "dry" and "wet" scenarios.

The cumulative funding required to offset these rates was calculated for the period 2022-28 (Figure 6). Results indicate that with a 5% annual rate increase limit, the reserve funds required would vary from \$112M for the wet scenario to \$148M for a dry supply scenario. Limiting annual rate increases to 10% would mean that CAP's reserve withdrawals would range from \$53M (wet) to \$65M (dry), and for a 15% limit, from \$23M (wet) to \$24M (dry).

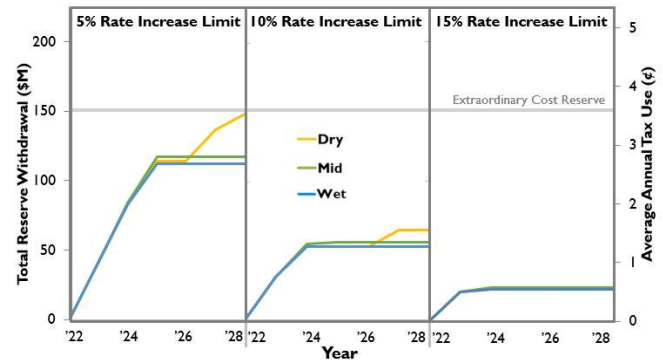


Fig. 6. Tradeoffs Between Rate Increases and Reserve Withdrawals

These values can be compared to the existing financial reserves that CAP maintains. The CAP's rate stabilization reserve maintains such a small balance (~\$22 million) that would be exhausted in all scenarios. The Extraordinary Cost Reserve (grey line in Figure 6) is designated as a resource meant to pay for large expenses that may cause unplanned "spikes" in the water rate, and currently maintains a balance of approximately \$150M. While these funds were not intended for rate stabilization, some portion could be used to effectively manage rates for the next 5 years under all scenarios.

Funds from the ad valorem tax could also be used, and these derive from an assessment of 10-cents per \$100 of valuation on properties within the CAP service area. On average, 1-cent of this tax brings in approximately \$7M in revenue per year. Some portion of the 10-cent tax could be combined with reserve funds to stabilize rates. The right-hand vertical axis in figure 6 describes the average annual tax use required over the 5-year period to manage rates in each scenario. While managing rates to a 10% or 15% annual rate increase limit would require using 1 to 2 cents of the average annual tax, maintaining a 5% rate increase limit would require approximately 3 cents, or 30%, of total ad valorem tax revenue.

V. DISCUSSION AND FUTURE WORK

This study describes the drought-related financial risk that large water providers face, as well as potential approaches to risk management that could promote climate resiliency for these operators of large-scale critical infrastructure. Reserve funds require constant recharge, but if properly managed, impose smaller and consistent rate increases on customers, preferable to the much larger and unpredictable increases that can arise due to drought. In this case, establishing a (much larger) rate stabilization fund would allow CAP to ensure financial stability for its customers who are not easily able to adapt their budgets

quickly from year to year. Alternatively, ad-valorem tax revenue could provide a constant source of revenue to assist in managing rates, but would require budgetary tradeoffs.

The next step in this analysis will be to evaluate various ways that a rate stabilization fund can be sustainably financed. In other words, how large would annual fund contributions need to be to provide an acceptable level of financial risk management for CAP's customers. Once the size of the fund is determined, the means of establishing it will be explored, which could include a small constant increase in volumetric rates (using excess revenues in wet periods to recharge the fund which has been drawn down in dry periods) or raising CAP's tax rates (a constant annual source of revenue), or some combination of both. The research will also look beyond the 5-year time horizon to examine possible long-term fluctuations.

Beyond these immediate tasks, future work will entail examining the effect of energy price volatility on CAP's financial resiliency. The Central Arizona Project is the largest electricity consumer in the state of Arizona, with 32% of its costs associated with power consumption. The cost of energy can swing widely, but CAP seems to have managed this risk reasonably well through a combination of diversification and timely trading. Nonetheless, there might be advantages to CAP investing in longer-term power purchasing agreements. Lastly, efforts will be undertaken to explore the financial impact of the federal policies under consideration to conserve water on the Colorado River, including the Bureau of Reclamation's recent initiative offering to pay Colorado River users up to \$250M to not use water. This work will explore the potential externalities of water conservation policies and provide a more robust understanding of the financial risk implications of both droughts and the institutional responses to them.

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REFERENCES

- [1] Baum, R., & Characklis, G. W. (2020). Mitigating drought-related financial risks for water utilities via integration of risk pooling and reinsurance. *Journal of Water Resources Planning and Management*, 146(6), 05020007.
- [2] Baum, R., Characklis, G. W., & Serre, M. L. (2018). Effects of geographic diversification on risk pooling to mitigate drought-related financial losses for water utilities. *Water Resources Research*, 54. <https://doi.org/10.1002/2017WR021468>.
- [3] BlackRock. (2021). Ninety-Five Percent of Global Insurers Believe Climate Risk is Investment Risk. <https://www.blackrock.com/corporate/newsroom/press-releases/article/corporate-one/press-releases/ninety-five-percent-of-global-insurers>.
- [4] Brajer, V., Church, L., & Cummings, R. (1989). The strengths and weaknesses of water scarcity and sovereignty. *Natural Resources Journal*, 29, 489–509.
- [5] Central Arizona Project (2021). 2022 2023 Biennial Budget. <https://library.cap-az.com/documents/departments/finance/2022-23-Biennial-Budget.pdf>.
- [6] Central Arizona Project (2022). CAP Water User Roundtable: Taxes, Capital Charges and Reserves. <https://cap-az.granicus.com/player/clip/621?&redirect=true&h=5d9829c9bf62a2f0bd755d372fcf620c>.
- [7] Central Arizona Project (n.d.). Fact Sheet: CAP Power. <https://library.cap-az.com/documents/factsheets/cap-fact-sheet-power-portfolio.pdf>.

- [8] Chaudhry, S., & Shrier, C. (2010). Energy sustainability in the water sector: challenges and opportunities. In *Proceedings of Annual Conference and Exposition* (pp. 20-24).
- [9] Cherchi, C., Badruzzaman, M., Oppenheimer, J., Bros, C. M., & Jacangelo, J. G. (2015). Energy and water quality management systems for water utility's operations: A review. *Journal of environmental management*, 153, 108-120.
- [10] Danilenko, A., Dickson, E., & Jacobsen, M. (2010). Climate change and urban water utilities: challenges and opportunities.
- [11] Fitch Ratings: Credit Ratings & Analysis for Financial Markets. (2023) from <https://www.fitchratings.com/research/corporate-finance/fitch-ratings-plans-enhanced-climate-risk-analysis-for-corporate-credit-ratings-15-02-2023>.
- [12] Gleick, P. H. (2003). Global freshwater resources: Soft-path solutions for the 21st century, *Science*, 302(5650), 1524–1528.
- [13] Hanemann, W. Michael. "Price and rate structures." *Urban water demand management and planning* 137 (1997).
- [14] Hughes, J. A., Tiger, M., Eskaf, S., Berahzer, S., Royster, I., Boyle, S., et al. (2014). *Defining a resilient business model for water utilities*. Denver, CO: Water Research Foundation.
- [15] IPCC (2022). *Climate Change 2022: Impacts, Adaptation and Vulnerability. Contribution of Working Group II to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change* [H.-O. Pörtner, D.C. Roberts, M. Tignor, E.S. Poloczanska, K. Mintenbeck, A. Alegría, M. Craig, S. Langsdorf, S. Löschke, V. Möller, A. Okem, B. Rama (eds.)]. Cambridge University Press. Cambridge University Press, Cambridge, UK and New York, NY, USA, 3056 pp., doi:10.1017/9781009325844.
- [16] Kasprzyk, J. R., P. M. Reed, B. R. Kirsch, and G. W. Characklis (2009), Managing population and drought risks using many-objective water portfolio planning under uncertainty, *Water Resour. Res.*, 45, W12401, doi:10.1029/2009WR008121.
- [17] Kundzewicz, Z., L. Mata, N. Arnell, P. Doll, P. Kabat, B. Jimenez, K. A. Miller, T. Oki, Z. Sen, and I. Shikomanov (2007), Freshwater resources and their management, in *Climate Change 2007: Impacts, Adaptation and Vulnerability. Contribution of Working Group II to the Fourth Assessment Report of the Intergovernmental Panel on Climate Change*, edited by M. Parry et al., pp. 173–210, Cambridge Univ. Press, Cambridge, U. K.
- [18] Leurig, S. (2010). The ripple effect: Water risk in the municipal bond market. <http://www.ceres.org/resources/reports/water-bonds>.
- [19] Mukherjee, S., & Mishra, A. K. (2021). Increase in compound drought and heatwaves in a warming world. *Geophysical Research Letters*, 48(1), e2020GL090617.
- [20] NYSERDA, New York State Energy Research & Development Authority. (2010). *Water & Wastewater Energy Management Best Practices Handbook*. NYSERDA. <http://www.nyserdera.ny.gov>.
- [21] Olmstead, S. M., & Stavins, R. N. (2009). Comparing price and nonprice approaches to urban water conservation. *Water Resources Research*, 45(4).
- [22] Pint, E. M. (1999). Household responses to increased water rates during the California.
- [23] The White House. (2021). *Executive Order on Climate-Related Financial Risk*. <https://www.whitehouse.gov/briefing-room/presidential-actions/2021/05/20/executive-order-on-climate-related-financial-risk/>.
- [24] Tiger, M., Hughes, J. A., & Eskaf, S. (2014). Designing water rate structures for conservation and revenue stability. Environmental Finance Center at the University of North Carolina at Chapel Hill. Chapel Hill, NC.
- [25] U.S. Bureau of Reclamation. (n.d.) Central Arizona Project. <https://www.usbr.gov/projects/index.php?id=504>.
- [26] U.S. Bureau of Reclamation. (2023). CRMS Ensemble Output, GitHub repository, <https://github.com/usbr/crmms/tree/dev/Output%20Data>.
- [27] Williams, A. P., Cook, B. I., & Smerdon, J. E. (2022). Rapid intensification of the emerging southwestern North American megadrought in 2020–2021. *Nature Climate Change*, 12(3), 232-234.
- [28] Zeff, H. B., & Characklis, G. W. (2013). Managing water utility financial risks through third-party index insurance contracts. *Water Resources Research*, 49, 4939–495